

C14-P19 • MAIN WHITE SHELL REVISION

15 receiving pilots: Ø1.6 → Ø1.9 mm; smooth, not threaded.

Reprint only the main white front for your chosen variant.

Black/red parts and the small white RF window are unchanged.

Black top G pilots stay Ø1.6; through-clearances stay Ø2.4.

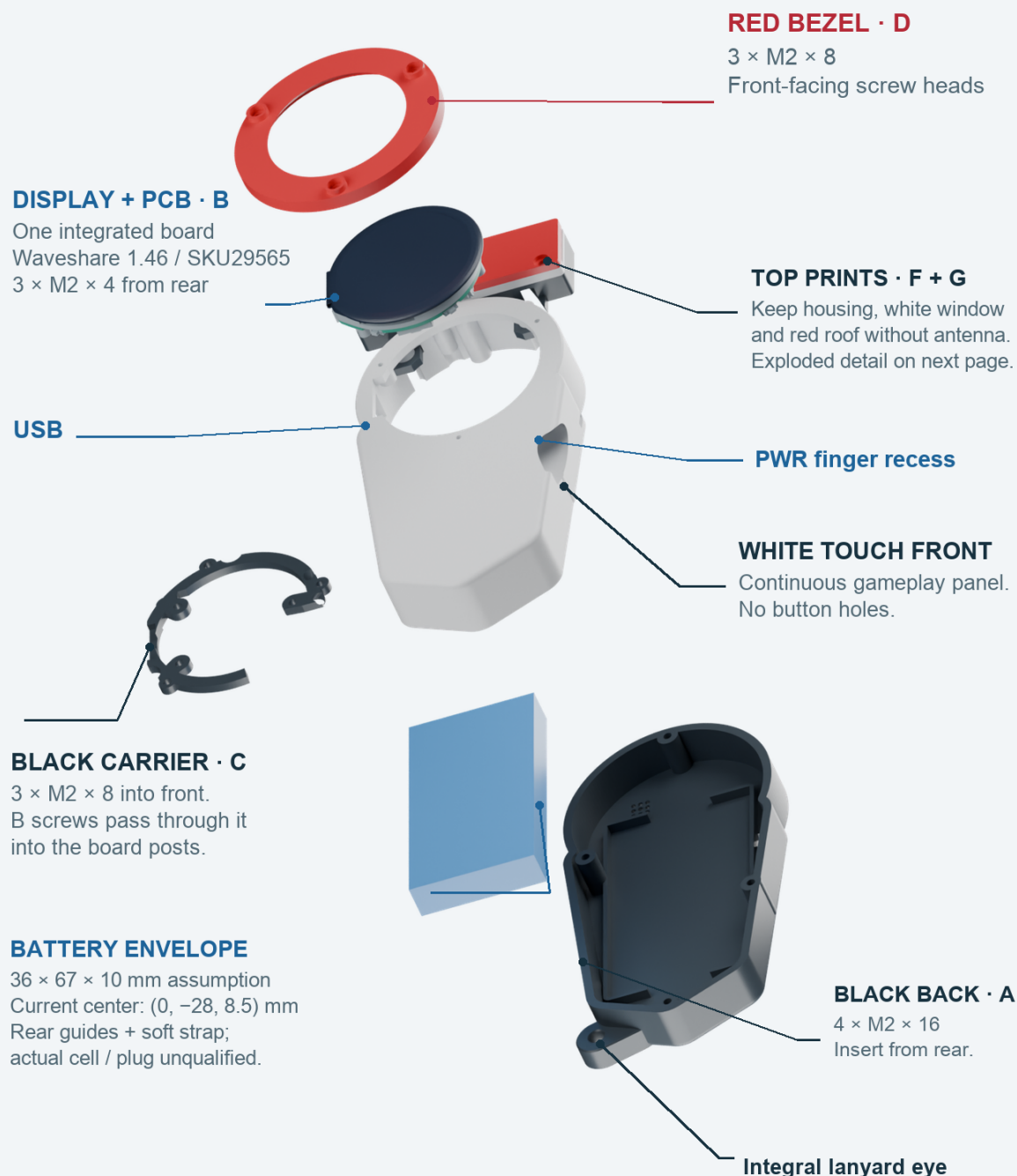
Larger pilots can reduce hold. Actual screw grip is untested.

CAD views reuse C13: outline and screw locations are unchanged.

z-ark Digivice • C14-P19

EXPLODED CORE / TOUCH_ONLY FIRST BUILD

NFC electronics absent • no external buttons or J9 harness



PART SEPARATION ≠ ASSEMBLY ORDER

Board enters from front with bezel off; carrier enters from rear.

Translations separate the actual meshes; orientation is preserved.

See screw maps for A–G positions. Nominal CAD fit; physical fit untested.

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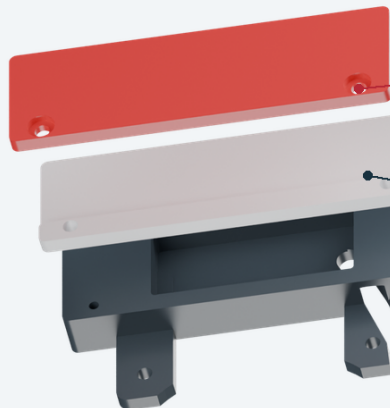
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z-ark Digivice • C14-P19 details

TOP PRINTS STAY IN THE FIRST BUILD

White RF window is required even when the NFC antenna is absent.



Card travel: across the top ↔

RED ROOF • G

2 × M2 × 8, from the front
through white window into housing

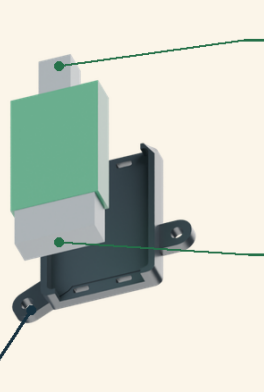
WHITE RF WINDOW

Captured beneath roof at screw strip.
No separate window screws.

BLACK HOUSING • F

2 × M2 × 8 from rear into front shell.
Leave antenna pocket empty for now.

LATER NFC OPTION — ABSENT / NOT ENABLED NOW



RF plug faces the top (+Y)

Host plug faces the bottom (-Y)



Antenna envelope
25 × 10 mm
Sits inside top housing

BLACK READER TRAY • E

3 × M2 × 8 into front shell.
Omit tray + E1–E3 for first build.

Reader orientation unchanged.
Center: Y = -42.5 mm.
Secure with soft straps;
E screws hold the tray,
not the reader PCB.

Colored electronics are
packing envelopes, not
detailed product shapes.

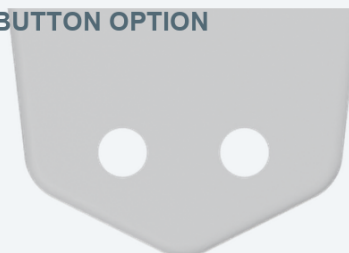
FRONT VARIANTS • SAME SCALE / LOWER FACE

TOUCH_ONLY



Solid face • no gameplay button hardware

TWO-BUTTON OPTION



Two holes • switch / harness fit unqualified

First build: 17 screws. With optional tray: 20 screws.

Actual hardware, battery, wire bends and physical assembly remain untested.

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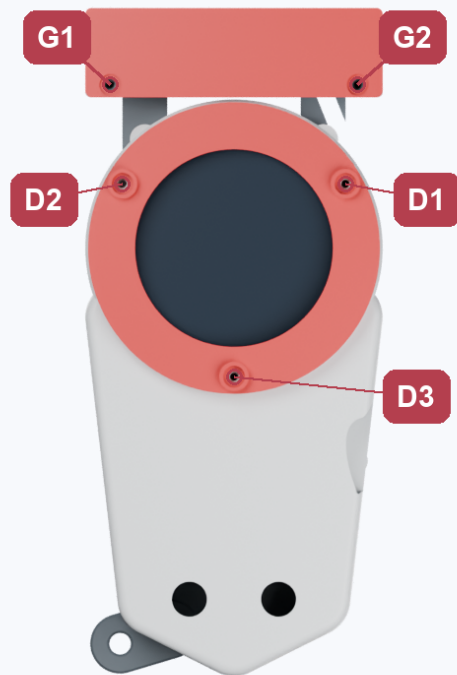
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01 EXTERIOR SCREWS

z-ark Digivice • C14-P19 • unchanged screw locations

FRONT

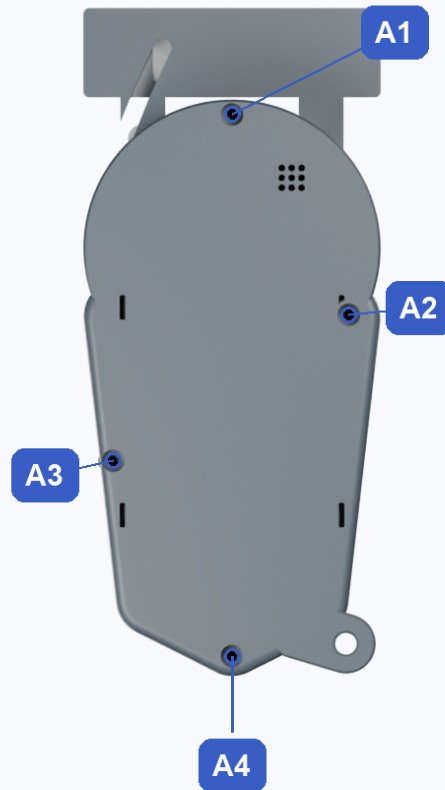
USB is on your LEFT



Button front shown
Touch-only closes the two holes.

REAR — CLOSED

USB is on your RIGHT



TOP SWIPE stays at the top in every view ↑

A1–A4

M2 × 16 mm • 4 screws

Back shell → front shell bosses

Insert from REAR; close the back last.

D1–D3

M2 × 8 mm • 3 screws

Red bezel → front shell bosses

Insert from FRONT; do not press on the glass.

G1–G2

M2 × 8 mm • 2 screws

Red roof + white RF window → black housing

Insert from FRONT. Window has no extra screws.

Both front variants use the same screw positions.

Button nuts/washers are separate, unqualified hardware.

Nominal CAD hardware • physical screw fit is untested.

C14-P19 • both variants • 1 device • 09 Oct 2026

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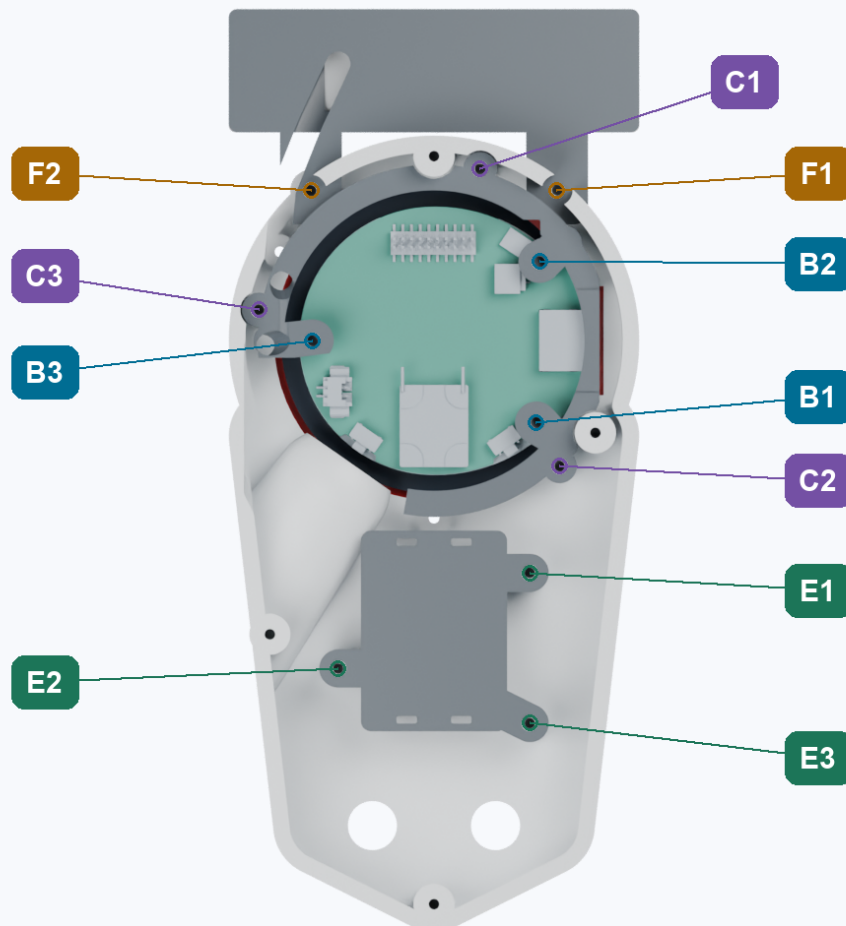
Black top G pilots stay Ø1.6; through-clearances stay Ø2.4.

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CAD views reuse C13: outline and screw locations are unchanged.

02 INTERNAL SCREWS

Open REAR view • USB on your RIGHT • top swipe ↑



BACK + BATTERY REMOVED FOR ACCESS

Reader absent here so tray mounting holes remain visible.

B1–B3

M2 × 4 mm · 3 screws

Carrier → board's three metal mounting posts

C1–C3

M2 × 8 mm · 3 screws

Carrier → front shell bosses

E1–E3

M2 × 8 mm · 3 optional screws

NFC reader tray → front shell bosses

F1–F2

M2 × 8 mm · 2 screws

Top black housing tabs → front shell bosses

No NFC: 17 screws. With NFC tray: 20 screws.

17 = 4 × 16 mm + 3 × 4 mm + 10 × 8 mm; add 3 × 8 mm.

Nominal CAD hardware • physical screw fit is untested.

C14-P19 • both variants • 1 device • 09 Oct 2026

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Larger pilots can reduce hold. Actual screw grip is untested.

Connection facts unchanged; the layout below is schematic.

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C14-P19 / how the electronics connect

FIRST BUILD: USB + integrated board; no J9 harness needed

USB cable

Power + data



Waveshare 1.46 / SKU29565

Screen + touch + ESP32-S3 are one unit.

No separate display or touch harness.

Onboard PWR: finger access; no extra wire.



J1 main battery

Battery connection is conditional

J1 is for nominal 3.7 V / 1S lithium. No pack is qualified yet.

Before connecting or charging, verify plug keying,
polarity and charger compatibility. Wire color is not proof.

Cell assumption: 36 × 67 × 10 mm; restrained in rear shell.

FUTURE WIRING / not enabled in current firmware

Button front: two normally-open momentary switches

Buttons A / B - - - - -> Finished harness - - - - -> J9: GPIO + GND

GPIO12 / GPIO13: future inputs. Touch-only omits buttons / wires.

Optional NFC: PN5321 MINI + remote top antenna

Board J9 - - - - -> Host harness - - - - -> PN5321 host port
7-pin MX1.25

PN5321 RF port - - -> 100 mm RF lead - - -> Top antenna

2-pin MX1.0 RF connection; separate from the host cable.

RF route unchanged: 87.1 mm + 10 mm planning allowance.

White RF window = plastic barrier, with no electrical wiring.

Future GPIO10/11 I²C + power/GND: harness still unqualified.

First build: no external buttons, NFC, GPS or J9 harness.

Layout is schematic; it does not show physical connector pins.

Sources: Waveshare 1.46 docs; Elechouse PN5321 MINI; inherited C13 facts.

Local firmware board profile checked 2026-10-09. Hardware fit remains untested.

Assembly order & screw schedule

C14-P19 / one device / touch-only first build without NFC

REPRINT ONLY THE MAIN WHITE FRONT for your chosen button or touch-only variant. All 15 white receiving pilots (A/C/D/E/F) change from diameter 1.6 to 1.9 mm. Black/red parts and the small white RF window stay unchanged. Smooth pilots are not threads; increased diameter can reduce screw hold. Physical grip remains untested.

- 1. Prepare.** Remove supports, especially from PWR and USB openings. Leave USB and battery disconnected. Check actual screw heads and trial-fit a spare 1.9 mm pilot with your screw. Do not force the hardware; there are no modeled threads, nuts or inserts in the white bosses.
- 2. Build the top.** Fit the black housing with **F1-F2, M2 x 8** from the rear. Place the white RF window, then red roof; **G1-G2, M2 x 8** capture both. Keep all three printed parts even without NFC.
- 3. Seat the screen/board.** Insert the integrated unit from the front with the red bezel removed. USB points left when viewing the screen. Fit the black carrier from the rear.
- 4. Secure the carrier.** Loosely start **B1-B3, M2 x 4** into the board posts and **C1-C3, M2 x 8** into front-shell bosses. Confirm flush alignment, then snug gradually. Do not pull a misaligned board into place with screws.
- 5. Finish the face.** Fit the red bezel using **D1-D3, M2 x 8** from the front. For this touch-only build, omit gameplay buttons and the NFC tray, reader, antenna and leads; leave **E1-E3** unused.
- 6. Check access and power.** Inspect PWR finger access and USB clearance before closure. Bring up the integrated board by USB with the battery unplugged. No J9 harness is needed for the first build.
- 7. Close last.** Only after battery plug/polarity/charger checks, insulate and softly restrain the cell in the rear shell. Battery position is unchanged from C13 (5 mm toward the screen versus the older layout). Keep wires clear, close without compression, and fit **A1-A4, M2 x 16** from the rear.

Nominal screw reach into the mating feature

IDs	Qty	Screw	From	Mating feature	Entry*
A1-A4	4	M2 x 16 mm	Rear	Back to front boss	6.4 mm
B1-B3	3	M2 x 4 mm	Rear	Carrier to metal board post	1.6 mm
C1-C3	3	M2 x 8 mm	Rear	Carrier to front boss	5.6 mm
D1-D3	3	M2 x 8 mm	Front	Bezel to front boss	4.5 mm
E1-E3	3	M2 x 8 mm	Rear	Optional tray to front boss	6.0 mm
F1-F2	2	M2 x 8 mm	Rear	Top housing to front boss	5.6 mm
G1-G2	2	M2 x 8 mm	Front	Roof + window to housing	4.2 mm

First build: 4 x M2x16 + 3 x M2x4 + 10 x M2x8 = **17**. Add the optional NFC tray and 3 x M2x8 (E) for **20**. Double these counts for two devices. Panel-switch mounting hardware is separate.

***Entry is a CAD axial overlap, not a tested thread engagement.** B reaches 1.6 mm into the nominal metal-post region; actual threads/blind depth must be checked. D3 has a rear-open boss, with 0.65 mm of target extending beyond the tip, not a closed floor.

Lengths are shaft lengths under the head. Head envelope: diameter 4 x 2 mm maximum. WHITE A/C/D/E/F pilots: diameter 1.9 mm; BLACK top G pilots remain diameter 1.6 mm; through-clearances remain diameter 2.4 mm. B mates to board metal posts. Actual head style, thread profile, grip and torque remain unqualified. Screw lengths, axes and pilot cut depths are unchanged. No M3 or extra window screws.